



**PAK-AUSTRIA FACHHOCHSCHULE:**  
**INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY**

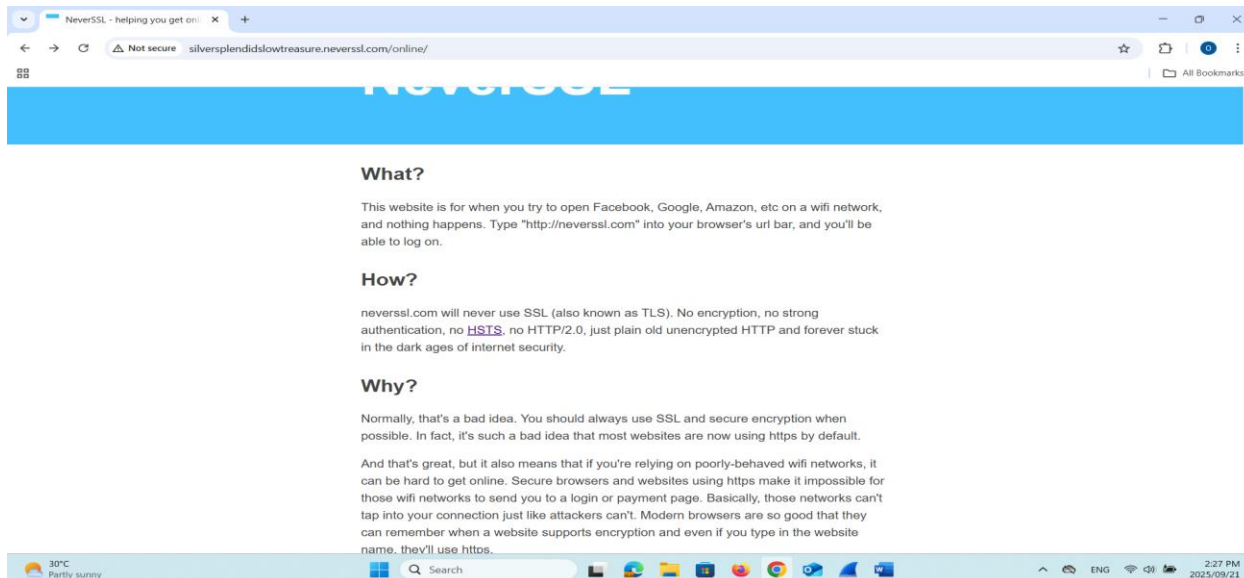
## **Computer Networks**

### **Assignment-01 Task#04**

|                                                                    |                                                                      |
|--------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>OSAMA MUHAMMAD</b><br><b>Amir Ayub</b><br><b>Abdullah Sajid</b> | <b>B23F0333AI079</b><br><b>B23F0767AI193</b><br><b>B23F0001AI062</b> |
| <b>DEPARTMENT</b>                                                  | <b>CS AND IT</b>                                                     |
| <b>PROGRAM</b>                                                     | <b>BS AI</b>                                                         |
| <b>SECTION</b>                                                     | <b>AI BLUE</b>                                                       |

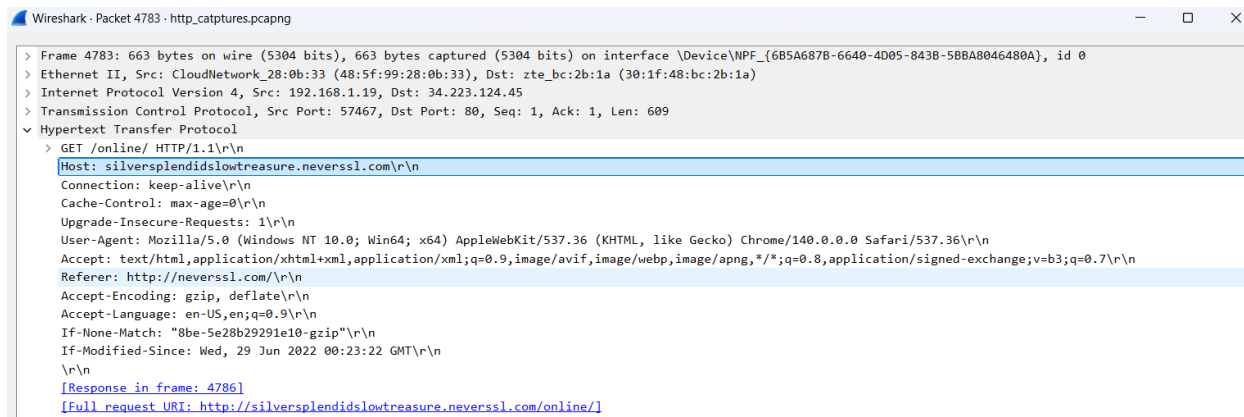
## Task:04

For the HTTP based website access, answer the following after analysing collected traces of HTTP:



### 1. What is the name of website?

Host:neverssl.com



### 2. Find the packet that contains the first GET request for the website you have accessed.

Packet number:4783

| http.request.method == "GET" |            |              |                |       |        |                                                                                                        |                                                                                          |
|------------------------------|------------|--------------|----------------|-------|--------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| No.                          | Time       | Source       | Destination    | Proto | Length | Full request URI                                                                                       | Info                                                                                     |
| 4783                         | 470.030797 | 192.168.1.19 | 34.223.124.45  | HT    | 663    | http://silversplendidslowtreasure.neverssl.com/online/                                                 | GET /online/ HTTP/1.1                                                                    |
| 6983                         | 802.469889 | 192.168.1.19 | 204.79.197.203 | HT    | 464    | http://oneocsp.com/ocsp/MFQwUjBQHE4wTDA3BglrDgKCGgUABBT0QYLFESE5G0K2FpaRVfYu7d9gZEBQAUZU2EpsABPY2zvadf | GET /ocsp/MFQwUjBQHE4wTDA3BglrDgKCGgUABBT0QYLFESE5G0K2FpaRVfYu7d9gZEBQAUZU2EpsABPY2zvadf |
| 7014                         | 802.534953 | 192.168.1.19 | 204.79.197.203 | HT    | 466    | http://oneocsp.com/ocsp/MFQwUjBQHE4wTDA3BglrDgKCGgUABBT0QYLFESE5G0K2FpaRVfYu7d9gZEBQAUZU2EpsABPY2zvadf | GET /ocsp/MFQwUjBQHE4wTDA3BglrDgKCGgUABBT0QYLFESE5G0K2FpaRVfYu7d9gZEBQAUZU2EpsABPY2zvadf |

### 3. Describe all headers and their values in this GET request message.

```
Hypertext Transfer Protocol
> GET /online/ HTTP/1.1\r\n
Host: silversplendidslowtreasure.neverssl.com\r\n
Connection: keep-alive\r\n
Cache-Control: max-age=0\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
Referer: http://neverssl.com/\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
If-None-Match: "8be-5e28b29291e10-gzip"\r\n
If-Modified-Since: Wed, 29 Jun 2022 00:23:22 GMT\r\n
\r\n
[Response in frame: 4786]
[Full request URI: http://silversplendidslowtreasure.neverssl.com/online/]
```

#### ❖ **GET /online/ HTTP/1.1:**

Method: GET → The client requests data from the server.

Resource Path: /online/ → The specific page or resource being requested.

Protocol Version: HTTP/1.1 → Communication is using HTTP version 1.1

#### ❖ **silversplendidslowtreasure.neverssl.com:**

Specifies the domain name of the server the request is being sent to.

Important in HTTP/1.1 because multiple domains can be hosted on the same IP

#### ❖ **keep-alive:**

Instructs the server to keep the TCP connection open for multiple requests/responses.

This supports **persistent connection**

#### ❖ **Cache-Control:**

Tells the server that cached content must not be older than 0 seconds.

In practice, it forces the request to check for fresh content on the server

#### ❖ **Upgrade-Insecure-Requests:**

Indicates the client prefers to upgrade from http:// to https:// if possible.

#### ❖ **Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36:**

Identifies the client software (web browser and OS).

Here: Chrome browser version 140 on Windows 10 (64-bit)

#### ❖ **Accept:**

Lists the content types the client is willing to accept.

text/html, application/xml, and several image formats (AVIF, WebP, APNG).

\*/\*;q=0.8 means it will accept any type with lower preference.

Quality factors (q=) represent priority

❖ **Referer:**

The page the request originated from.

Indicates the client navigated to this resource from the neverss1.com domain

❖ **Accept-Encoding:**

Specifies the compression methods the client can handle.

Server may send compressed data to reduce size

❖ **Accept-Language:**

Preferred languages for the response.

English (United States) is top priority, followed by general English

❖ **If-None-Match:**

Contains an ETag value (entity tag).

Client asks the server: “Send me the resource only if it has changed since this version.”

Used for caching and reducing unnecessary downloads

❖ **If-Modified-Since:**

Another caching mechanism.

Client asks the server to send the file only if it has been modified after this date.

❖ **Full Request URI:**

This is the complete URL being requested

#### 4. Identify the status code in the first server response.

Status Code:200 OK

| http.response |            |                |              |       |       |                    |                                        |
|---------------|------------|----------------|--------------|-------|-------|--------------------|----------------------------------------|
| No.           | Time       | Source         | Destination  | Prot  | Leng  | Full request URI   | Info                                   |
| 4786          | 470.410253 | 34.223.124.45  | 192.168.1.19 | HT... | 14... | http://silversp... | HTTP/1.1 200 OK (text/html)            |
| 7001          | 802.523312 | 204.79.197.203 | 192.168.1.19 | OC... | 29... | http://oneocsp...  | Response[BoundErrorUnresembled Packet] |
| 7046          | 802.597099 | 204.79.197.203 | 192.168.1.19 | OC... | 29... | http://oneocsp...  | Response[BoundErrorUnresembled Packet] |

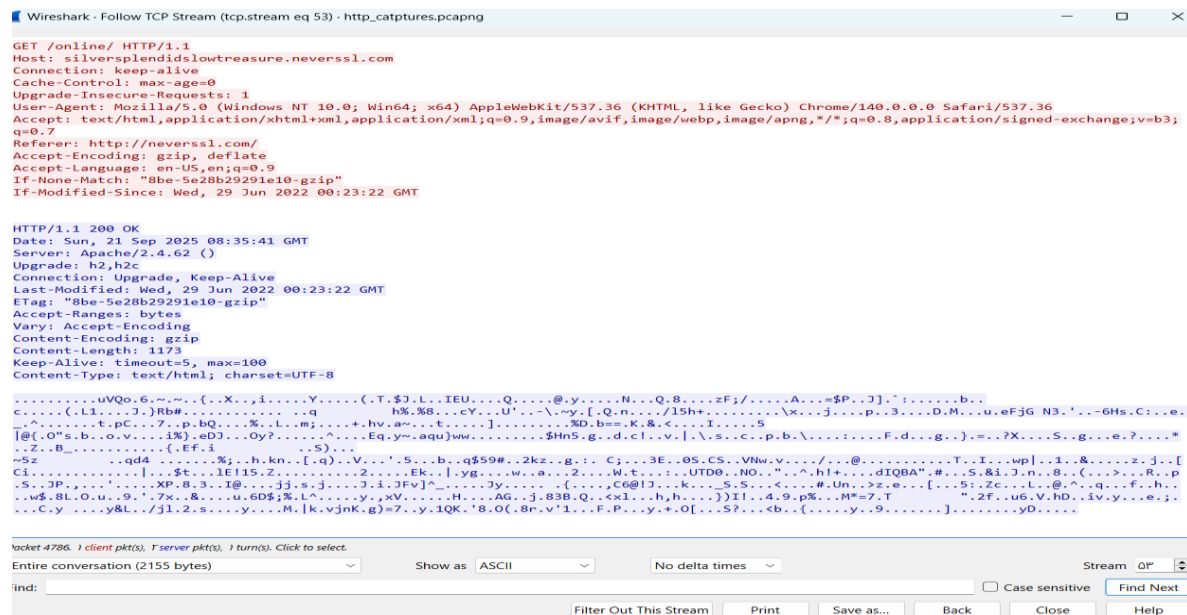
#### 5. How many HTTP response messages are exchanged in total?

Total number of packets exchanged:3

| http.response |            |                |              |       |       |                    |                                        |
|---------------|------------|----------------|--------------|-------|-------|--------------------|----------------------------------------|
| No.           | Time       | Source         | Destination  | Prot  | Leng  | Full request URI   | Info                                   |
| 4786          | 470.410253 | 34.223.124.45  | 192.168.1.19 | HT... | 14... | http://silversp... | HTTP/1.1 200 OK (text/html)            |
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| 7046          | 802.597099 | 204.79.197.203 | 192.168.1.19 | OC... | 29... | http://oneocsp...  | Response[BoundErrorUnresembled Packet] |

## 6. Determine whether the connection is persistent or not. Justify with evidence from packet captures.

The connection is Persistent because the client-server connection contain the keep-alive connection header.



Wireshark · Follow TCP Stream (tcp.stream eq 53) · http\_captures.pcapng

```
GET /online/ HTTP/1.1
Host: silversplendidslowtreasure.neverssl.com
Connection: keep-alive
Cache-Control: max-age=0
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9
If-None-Match: "8be-5e28b29291e10-gzip"
If-Modified-Since: Wed, 29 Jun 2022 00:23:22 GMT

HTTP/1.1 200 OK
Date: Sun, 21 Sep 2025 08:35:41 GMT
Server: Apache/2.4.62 ()
Upgrade: h2,h2c
Connection: Upgrade, Keep-Alive
Last-Modified: Wed, 29 Jun 2022 00:23:22 GMT
ETag: "8be-5e28b29291e10-gzip"
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Encoding: gzip
Content-Length: 1173
Keep-Alive: timeout=5, max=100
Content-Type: text/html; charset=UTF-8

.....uVQ0.6~..{..X..i.....Y.....(.T.$J.L.IEU...Q....@.y....N...Q.8....zF;.....A...=$P..J].~i.....b..
c....(.L1....J.)Rb#.....q.....h%.%8...cY...U'...~\..my.{.Q.n....../15h+.....\X...J....p..3....D.M...u.eFjG N3.'...6Hs.C:...e.
_~^.....t.pC...7...p.bQ....%.L..m;.....+hv.a~...t....].....%D.b==K.&.<...I.....S
|@{.'0's.b..o.v....i%}.eDj...Oy?.....^.....Eq.y~.aqu)ww.....$Hn5.g...d.cl..v.{.\s...c..p.b.\.....F.d...g...}.~..?X...S...g...e.?....*
~Z..B.....{..Ef.i.....S)...
~qd4 .....%;..h.kn..[.q)..V...'.5...b..q$59#.2kz..g.:..Cj...3E...0S.CS..VNw.v....../...@.....T..I...wp[...1..&.....z..j..[
Ci.....[...$.t...IE115.Z.....2....Ek..|.yg...w...a...2....W.t....UTD0..NO..".^..hl+...dIQBA".#...S.&i.J.n..8...(>...>...R...p
$.3P.....XP..8.3..I0...~jj.s-j....J.i.3Fv|^.....Jy:....{.....C60Ij...k....S_S...<...#Um...>z.e...[...5z.Zc...L..@^..q...f..h...
..w$.8L.O.u..9.'7x..&...u.6D$;%L^.....y..xv.....H....AG..j.83B.Q.<x1..h,h....})I...4.9.p%..M*=7.T...".2F...u6.V.hD...iv.y...e..j...
...C.y...y&L../j1.2.s...y...M.[k.vjnK.g)=7..y.1QK.'8.O(.8r.v'1...F.P...y.+O[...S?...<b...{.....y..9.....}.....yD.....
```

Packet 4786: 1 client pkt(0), 1 server pkt(0), 1 turn(0). Click to select.

Entire conversation (2155 bytes) Show as ASCII No delta times Stream 0F

Find Next

Filter Out This Stream Print Save as... Back Close Help